

Atta Ul Asad

M.S. Student in Artificial Intelligence, Lahore University of Management Sciences (LUMS)

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RESEARCH INTERESTS

Clinical AI and Human–AI Decision-Making. Studying LLMs as clinical decision support rather than as standalone diagnosticians. My current work measures second-opinion independence: whether showing a model the clinician’s working diagnosis suppresses its ability to flag that diagnosis as an error, using documented real versus synthetic misdiagnoses. I am interested in anchoring, automation bias, and sycophancy toward incorrect clinical reasoning, and in protocols that keep a second opinion independent.

Trustworthy LLM Reasoning. Auditing failures that aggregate accuracy hides. Under INT4 KV-cache quantization, over 90% of faithfulness changes among RAG answers that stay correct are negative; my audits also test whether a decoding-time penalty beats generic repetition control on quantization-induced reasoning loops, and whether per-agent unlearning survives multi-agent re-derivation through shared memory. I design these studies around paired or clustered statistics and explicit failure criteria.

Efficient Inference and Self-Improving Agents. Designing training-free KV-cache compression that treats eviction as predicting which cached tokens future decoding queries will need, using a small, bounded amount of test-time compute on sampled continuations. I also study coding agents that improve through reinforcement learning on their own trajectories, and when that self-training yields genuine gains rather than reinforcing the agent’s own errors.

EDUCATION

Lahore University of Management Sciences (LUMS)

August 2026 – Present

Master of Science in Artificial Intelligence

- **Coursework** Foundations of AI, Mathematics for AI

COMSATS University Islamabad

September 2022 – June 2026

Bachelor of Science in Computer Science

- **Dean’s List of Honors** SGPA 3.98/4.00 in the third semester
- **Senior Design Thesis** AI Teaching Assistant System: A Multimodal Intelligent Assistant for Automated Student Support and Analytics
- **Relevant Coursework** Natural Language Processing, Machine Learning, Data Structures and Algorithms, Database Systems, Linear Algebra, Object-Oriented Programming

RESEARCH EXPERIENCE

Lahore University of Management Sciences (LUMS), Independent Researcher

August 2026 – Present

Project: Second-Opinion Independence in Clinical LLM Decision Support

(in progress, ACM FAccT 2027)

- Studying whether a clinician’s working diagnosis suppresses an LLM’s ability to detect the error when the model is used as a second opinion, comparing real documented misdiagnoses against synthetic errors and measuring change in error detection rather than standalone accuracy.
- Designing a judge-free evaluation on 300 of the 897 MedCaseReasoning test cases with Qwen3-8B served through vLLM, constrained decoding to a frozen diagnosis vocabulary, exact-match scoring, and a case-level paired bootstrap (10,000 resamples, 95% CIs).

Stanford AI Lab (SAIL), Stanford University, Research Collaborator

July 2026 – Present

Collaborators: *M. A. Mohsin* and *M. Umer*

- **LORE-KV(ICLR 2027)**. Co-authored a training-free KV-cache eviction method that samples short continuations from the frozen target model and scores each prompt token by its projected leave-one-out attention-output deletion cost, aggregated across sampled futures. At a 128-token budget it raised LongBench on Qwen2.5-14B from 45.49 to 48.24 and 16K RULER on Mistral-7B from 45.20 to 51.05; one sampled continuation recovers about 89% of the gain over the prompt-window control. Joint work with researchers at the University of Oklahoma, the University of Cambridge, the University of Houston, and Isara Labs.
- **Self-Improving Coding Agents (in progress, ICML 2027)**. Studying whether a coding agent built on Qwen2.5-14B can improve through reinforcement learning on its own sampled trajectories, evaluated on HumanEval+ and MBPP+. Built the agent scaffold and the evaluation harness that executes generated solutions against the EvalPlus test suites.

University of Oklahoma, Research Intern

June 2026 – August 2026

Advisor: *Dr. Dean F. Hougen* | Project: *Faithfulness Is Not Free (GroundLM @ EMNLP 2026)*

- **KV-Cache Quantization in RAG**. Audited BF16, INT8, and INT4 quantization of offline document KV caches in RAG with Qwen2.5-7B-Instruct on RGB and HotpotQA, measuring accuracy (EM, F1) and faithfulness with HHEM, DeBERTa-v3 NLI, and an LLM judge.
- Showed that INT8 is near-lossless while INT4 reduces accuracy and, among answers that remain correct, over 90% of faithfulness changes are negative, a regression accuracy metrics miss; the harm grows under noisy retrieval and with more chunks (paired McNemar tests, OLS).

University of Malaya, Undergraduate Research Assistant

July 2025 – September 2025

Advisor: *Dr. Uzair Ishtiaq*

- Built BioBERT-based clinical NLP pipelines for biomedical named-entity recognition, clinical text classification, and disease-related information extraction.

PUBLICATIONS

* denotes equal contribution. Author name in **bold**.

Conference Submissions

[C1] Monte Carlo Future-Query Estimation for KV Cache Eviction

ICLR 2027

A. Bilal, M. A. Mohsin, M. Umer, W. H. Raza, A. U. Asad, Y. D. Kwon, M. Valko, D. F. Hougen

Workshop Papers

[W1] Faithfulness Is Not Free: Auditing Offline KV-Cache Quantization in Retrieval-Augmented Generation

A. U. Asad, A. Bilal*, M. Ali, M. Haseeb, D. F. Hougen [arXiv]*

RESEARCH PROJECTS

Quantized Self-Correction Audit

[Code](#)

Preregistered study | Python, PyTorch, vLLM, AWQ

- Tested whether the “overthinking” marker penalty of Lotfi et al. (2026) fixes the looping that 3-bit AWQ induces in DeepSeek-R1-Distill-Qwen-1.5B on MATH-500 (500 paired traces, five arms); quantization alone raised the loop rate by 0.56 and cut accuracy from 0.85 to 0.39.
- The penalty was statistically indistinguishable from a one-line repetition penalty on loop rate and accuracy, and reaches only 49 of its 50 markers and 42.9% of probed contexts.

Re-derivation Leakage in Multi-Agent Unlearning

[Code](#)

Preregistered study | Python, PyTorch, open-unlearning, TOFU

- Showed that shared memory enforcing a retrieval-time ID blacklist and deletion over the derivation closure can still store forgotten TOFU facts: a node written from an agent’s parametric answer has a fresh ID and no parent edges, so both invariants certify it clean.

- With NPO forget10 agents, realistic routing delegated only 3 of 400 forget-set questions.

Natural Language Processing

[Code](#)

Python, PyTorch, HuggingFace Transformers, FAISS, NLTK

- **Cross-Lingual Sentiment Analysis.** Fine-tuned XLM-RoBERTa on Urdu–English code-mixed Twitter data, outperforming mBERT by 4.2% F1.
- **Biomedical NER.** Fine-tuned BioBERT on the NCBI Disease corpus with precision and recall analysis.
- **Dense Passage Retrieval.** Built a DPR and FAISS open-domain QA pipeline evaluated on Natural Questions.

Large Language Models

[Code](#)

Python, PEFT, vLLM, LangChain, Ollama

- **LoRA Fine-Tuning.** Applied 4-bit LoRA to LLaMA-2-7B on a single GPU and analyzed rank-wise representational change with CKA and linear probes.
- **KV-Cache Benchmarking.** Compared eviction and quantization policies on LLaMA-3 and Mistral-7B for latency, memory footprint, and generation quality.
- **RAG Faithfulness.** Built an NLI-based hallucination detector for RAG outputs.

TEACHING EXPERIENCE

COMSATS University Islamabad, Undergraduate Teaching Assistant Spring 2024 – Fall 2025

- **Machine Learning** (Fall 2025), **Natural Language Processing** (Spring 2025): Conducted weekly lab sessions and tutored 100+ students.
- **Data Structures & Algorithms** (Fall 2024), **Object-Oriented Programming** (Spring 2024): Conducted lab demonstrations, evaluated course projects, and held office hours.

INDUSTRY EXPERIENCE

King Revolution (Pvt.) Ltd., Machine Learning Intern January 2026 – March 2026

- Built end-to-end supervised and unsupervised ML pipelines, from data acquisition and feature engineering to deployment.

FFC, Data Analyst Intern June 2024 – August 2024

- Analyzed operational datasets to identify trends, patterns, and insights for business processes.
- Performed data cleaning, preprocessing, and exploratory data analysis (EDA).

REVIEWER AND SERVICE

Workshop Reviewer GroundLM @ EMNLP 2026, VLM4RWD 2026

Student Leadership Management Lead, IEEE Computer Society, COMSATS University Islamabad

TECHNICAL SKILLS

Languages Python, C++, SQL, JavaScript, $\text{\LaTeX}$

ML and LLMs PyTorch, HuggingFace Transformers, PEFT, vLLM, TensorFlow, scikit-learn, FAISS

Systems Git, Docker, Linux, FastAPI, Django, PostgreSQL, MongoDB

HONORS AND ACHIEVEMENTS

Harvard CS50 Puzzle Day 2026 Solved all 10 puzzles ([certificate](#)).

MIT Informatics Program Participant ([certificate](#)).

Competitions Meta Hacker Cup, AMD Developer Hackathon, UC Berkeley Competition.

CERTIFICATIONS

Coursera [Machine Learning Specialization](#) (DeepLearning.AI and Stanford Online), [Deep Learning Specialization](#) (DeepLearning.AI), [Google Python Programming Certificate](#), [Data Analysis with Python](#) (University of Pennsylvania), [Django Web Development](#) (Meta).